# Kiran Yamanappa Alur

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## Summary

Electronics and Communication Engineering student with a strong interest in digital systems and embedded technologies. Gained hands-on experience in Verilog HDL, FPGA design, embedded systems, C/C++, and Python through academic and internship projects Strong analytical and problem-solving skills, with the ability to collaborate effectively in team environments and adapt quickly to new technologies. Passionate about developing practical engineering solutions and eager to contribute to innovative projects while continuously expanding technical expertise.

## Skills

**Languages:** C, C++, Python | **HDL:** Verilog HDL | **Embedded Systems:** Arduino, ESP32| **Tools:** Vivado, Vitis Model Composer, MATLAB/Simulink, ModelSim / QuestaSim, Git, VS Code, Antigravity, Andriod Studio | **Concepts:** Digital Electronics, Computer Architecture, Data Structures & Algorithms (C++), Operating Systems.

## Experience

**Summer Intern**: **Crofting Technologies** ( 01/07/26 to 12/08/26 ) [ Completion Certificate ]

- Designing and implementing a **Verilog HDL-based RISC-V processor integrated with an Ethernet communication interface**, performing RTL development, functional simulation, debugging, and FPGA validation using Xilinx Vivado and Vitis Model Composer. Achieved 84% code coverage for the Ethernet module using ModelSim/QuestaSim. [ Link ]
- Developing **an FPGA-based object detection system** by implementing image processing algorithms, validating them using MATLAB/Simulink, optimizing RTL, and resolving hardware integration issues. Achieved 93% detection accuracy and 78% precision during system evaluation. [ Link ]

## Projects

**Women's Safety System with Integrated Mobile Application**

- Designed and developed a women's safety system capable of sending emergency alerts with the user's live location. Validated the system by conducting integration testing across 20 different locations and 30 user accounts, verifying reliable synchronization between the Android application and web portal while ensuring robust event handling and system reliability. Technologies: Android Studio, HTML, CSS, Node.js

**Air Mouse Using ESP32**

- Developed a wireless air mouse using ESP32 and the MPU6050 motion sensor. Implemented Bluetooth Low Energy (BLE) communication to control the computer cursor without additional drivers.
- Developed a low-cost assistive computer-interaction prototype using accelerometer and gyroscope data for smooth cursor movement and gesture-based control, optimized through iterative testing and calibration.

## Certifications

**Programming in C – Wingspan (Aug 2024)**: Completed practical training in C programming, including control structures, arrays, pointers, functions, and structures.

**Digital Electronics – Online Digital Electronics Course (June 2026)**: Completed training in digital logic, electronic circuits, and fundamental digital system design.

**VLSI Digital Design – Chip Design & Verilog Programming (July 2026)**: Completed training in VLSI digital design, chip design concepts, and Verilog programming.

**VLSI Workshop – ChipXpert Technologies Pvt. Ltd.(May 2026)**: Gained hands-on knowledge of RTL design, Verilog programming, digital design concepts, and VLSI design flow.

## Education

| Qualification | Institution | Year of passing | Score |
|---|---|---|---|
| B.E. (ECE) | SDM College of Engineering & Technology, Dharwad | 2027* | 9.38(up to 6th Semester) |
| PUC (Science) | Empower PU Science College, Dharwad | 2023 | 91% |
| SSLC | BGS High School, Dharwad | 2021 | 96% |